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TOPICAL REVIEW

Content Advisor: Johannes Boltze, MD, PhD

Consensus Review of Perioperative Immune Modulation for Endovascular Thrombectomy Following Ischemic Stroke

Ling Cai, MD*; Yan Li, MD, PhD*; Chong Wang, PhD; Tim Magnus, MD, PhD; Chengye Yao, MD, PhD; Xiaoming Hu, MD, PhD; Ann M. Stowe, PhD; Anna Rosell, PhD; Sunghee Cho, PhD; Stuart M. Allan, PhD; Jieqing Wan, MD, PhD; Yulong Ma, MD, PhD; Yang Liu, MD, PhD; Daniel I. Sessler, MD; Johannes Boltze, MD, PhD; Tianlong Wang, MD, PhD; Weifeng Yu, MD, PhD; Yueman Zhang, MD, PhD; Peiying Li, MD, PhD

ABSTRACT: Ischemic stroke is a major global health burden, leading to considerable mortality and long-term disability. Endovascular thrombectomy and mechanical recanalization have revolutionized acute stroke care. Nonetheless, many patients experience poor long-term neurological outcomes, which are often attributed to the no-reflow phenomenon and activation of inflammatory cascades. The perioperative period of endovascular thrombectomy, managed under either general anesthesia or conscious sedation, represents a critical window where anesthetic strategies may influence recovery through hemodynamic control and possibly immune modulation. This consensus review was generated by an international multidisciplinary expert group and synthesizes preclinical and clinical evidence to evaluate the promise of various immunomodulatory strategies for improving functional outcomes in patients with ischemic stroke following endovascular thrombectomy. Our goal is to provide a foundational reference for future research and development of novel perioperative immune therapies for patients with endovascular thrombectomy.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

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Key Words: anesthesia ■ conscious sedation ■ hemodynamics ■ immunology ■ stroke ■ thrombectomy

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Ischemic stroke remains a leading global cause of mortality and disability,[1] accounting for 70% of all strokes and causing 12% of global fatalities.[2] Over the past 3 decades, the global incidence of stroke increased by 70%, accompanied by a 43% increase in stroke-related mortality.[3] Improved acute stroke care increases the survival of patients with stroke but often experiences disability and dependency for years or even decades.[4] Cost-of-illness analyses estimate that the financial burden of strokes reaches up to US $100 000 per case, imposing a substantial socioeconomic impact.[5]

Endovascular thrombectomy (EVT) reduces the risk of 90-day mortality in patients with acute ischemic stroke by 3.7%.[6] When combined with intravenous thrombolysis, EVT improves outcomes compared with thrombolysis alone.[7] Based on these clinical benefits, current guidelines endorse EVT for patients with large-vessel occlusion, severe neurological deficits, and smaller infarct cores if treatment can be initiated within 6 hours of symptom onset.[8] Selected patients with Alberta Stroke Program Early Computed Tomography Scores between 3 and 5 within 6 to 24 hours after symptom onset may

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Correspondence to: Yueman Zhang, MD, PhD, Department of Anesthesiology, Renji Hospital and Key Laboratory of Anesthesiology, Shanghai Jiao Tong University, Ministry of Education, Shanghai Jiao Tong University School of Medicine, 160 Pujian Rd, Pudong New Area, Shanghai 200127, China, Email zhangyueman@renji.com; or Peiying Li, MD, PhD, Department of Anesthesiology, Renji Hospital and Key Laboratory of Anesthesiology, Shanghai Jiao Tong University, Ministry of Education, Shanghai Jiao Tong University School of Medicine, 160 Pujian Rd, Pudong New Area, Shanghai 200127, China, Email peiyingli.md@gmail.com

*L. Cai and Y. Li contributed equally.

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also benefit from revascularization.[9] There was a 3-fold increase in EVT use globally during the initial 5 years of clinical adoption from 2015 to 2020.[10] Unsurprisingly, more frequent EVT comparably increased the need for anesthesia.

EVT is usually performed with general anesthesia or conscious sedation, with selection guided by patient-specific factors including cooperation and neurological status.[11] The hope is that optimal anesthetic management during this critical phase might improve long-term neurological recovery by maintaining hemodynamic stability and possibly reducing activation of harmful inflammatory cascades.[12]

Many patients with ischemic stroke suffer long-term neurological sequelae despite successful mechanical recanalization. A major reason is failure of small vessel reperfusion despite large-vessel recanalization, termed the no-reflow phenomenon.[13] Poststroke neuroinflammatory cascade also contributes to poor neurological outcomes.[14] Emerging evidence suggests that perioperative immune modulation is a promising strategy to improve stroke outcome in patients with EVT.[15,16] Here, an international multidisciplinary expert group provides a consensus review to summarize the preclinical evidence and available trials of immunomodulatory approaches for ischemic stroke including cytokine-targeted therapies, immune cell migration inhibitors, anti-inflammatory agents, and cell-based therapies.

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THE RATIONALE OF PERIOPERATIVE IMMUNE MODULATION FOR PATIENTS WITH STROKE UNDERGOING EVT

There is a dynamic interplay between the central nervous system (CNS) and the immune system after strokes.[17] For example, elevated peripheral inflammatory markers such as IL (interleukin)-6, IL-1β, and TNF-α (tumor necrosis factor alpha) are poor prognosis signs. EVT, therefore, defines a critical window for combined anesthetic and immunomodulatory interventions.[18]

The ischemic brain exhibits temporally defined immune responses: initial innate activation (neutrophils, microglia, and macrophages) within hours to days,[19] followed by adaptive lymphocyte responses.[17] Spatiotemporal immune heterogeneity can be both a prognostic biomarker and a therapeutic target. Perioperative immune modulation is particularly relevant in this context: while surgical trauma may exacerbate inflammation, precisely timed immunotherapy may possibly confer cerebroprotection.[20] Poststroke neuroimmune interactions and EVT-induced inflammation represent a critical perioperative window where targeted immune modulation may improve outcomes in patients with stroke having EVT.

PERIOPERATIVE IMMUNOLOGIC CONSIDERATIONS AND PRECLINICAL ADVANCES FOR PATIENTS WITH ISCHEMIC STROKE

Immunologic Changes Following Ischemic Stroke

Ischemic stroke induces dynamic immunologic responses that are associated with worse immediate EVT outcomes and postprocedural recovery. The hyperacute phase (0–6 hours) represents the optimal period to revascularize while minimizing reperfusion injury risks (Figure 1).[17] Different phases demonstrate unique immunologic signatures, necessitating comprehensive characterization of stage-specific immune mechanisms.[21] Precise understanding of these phase-specific immune mechanisms provides the biological rationale for optimizing anesthetic management and adjunctive immunomodulatory strategies during EVT.[22]

Acute Neuroinflammation Following Ischemic Stroke

A serious complication during the acute phase following EVT is hemorrhagic transformation, a potentially devastating process arising from blood-brain barrier (BBB) disruption associated with acute neuroinflammation. The ischemic injury triggers rapid activation of resident glial cells and subsequent peripheral immune cell infiltration, establishing a destructive inflammatory cascade.[23] Microglia activation and astrocyte reactivity lead to cytokine release and MMP (matrix metalloproteinase)-2/9 secretion, which induces BBB leakage by degrading tight junctions and basement membranes.[24]

This dual glial response to hemorrhagic transformation establishes a chemotactic gradient that facilitates robust infiltration of immune cells from the periphery.[17] Upon activation, recruited peripheral immune cells, particularly neutrophils, exacerbate neuroinflammation by releasing cytotoxic granules and stimulating production of reactive oxygen and nitrogen species, which amplify CNS inflammation.[25] The interplay between CNS-resident glial cells and peripheral immune infiltration drives neuroinflammation and BBB disruption in the first hours to days after ischemic stroke, increasing hemorrhagic transformation risk and worsening outcomes. Modulating key immune signaling molecules that drive BBB disruption during the acute phase of stroke may, therefore, prove beneficial.

Stroke-Induced Acute Peripheral Immune Dysregulation Exacerbates Brain Injury and Affects Long-Term Outcomes

The ischemic brain engages in bidirectional communication with the peripheral immune system through multiple pathways.[26] Within hours of onset, sympathetic activation

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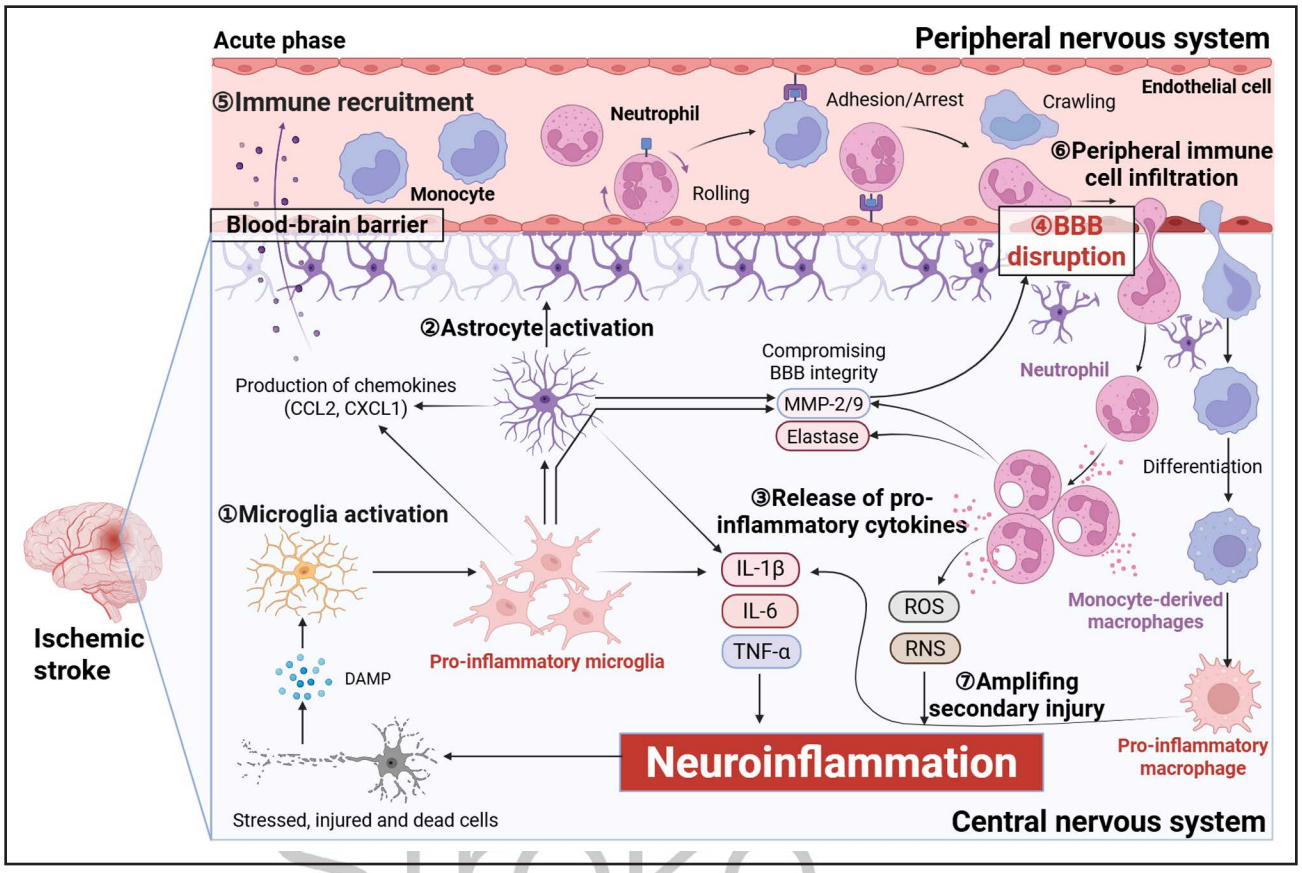

Figure 1. Mechanisms of central and peripheral immune responses in acute ischemic stroke.
Following ischemic stroke, microglia and astrocytes are promptly activated, releasing proinflammatory mediators and MMPs (matrix metalloproteinases) that compromise blood-brain barrier (BBB) integrity. Neutrophils subsequently infiltrate the parenchyma and exacerbate inflammatory damage through the secretion of cytotoxic agents. This is followed by the recruitment of monocyte-derived macrophages into the brain, which perpetuate the immune response. Collectively, these processes form a self-sustaining vicious cycle that amplifies cerebral injury and promotes systemic immunosuppression. DAMP indicates damage-associated molecular pattern; IL, interleukin; RNS, reactive nitrogen species; ROS, reactive oxygen species; and TNF-α, tumor necrosis factor alpha. Created in BioRender. Yan, L. (2026) https://BioRender.com/mn8arn4.

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induces catecholamine-mediated splenic contraction, mobilizing monocytes, neutrophils, and lymphocytes,[27] while circulating damage-associated molecular patterns amplify systemic inflammation.[28] Stroke-induced neuroinflammatory response is followed by a peripheral CNS-injury–induced immunosuppressive syndrome[29] that is characterized by lymphopenia and an immunosuppressive milieu that predisposes patients to secondary infections.[30]

The underlying mechanisms include sympathetic-mediated splenic atrophy,[31] glucocorticoid-induced myeloid suppression,[32] and damage-associated molecular pattern–driven expansion of immunosuppressive myeloid cells.[33] About one-third of patients with EVT develop peripheral infections within 7 days after a stroke,[34] which exacerbate cerebral injury through multiple pathways, including systemic cytokine release, endothelial dysfunction, and dysregulated coagulopathy.[35,36]

Collectively, stroke-induced immunosuppression increases infection risk, while subsequent infections can amplify neuroinflammation and impair long-term neurological recovery, thus forming a vicious cycle between ischemic brain injury and immunosuppression after stroke.

Immune Phenotype Transition and Biphasic Regulation in the Subacute Phase of Stroke

The subacute phase (3–7 days post-ischemia) is characterized by immunologic processes that may influence poststroke recovery (Figure 2). During this period, the inflammatory environment transforms as microglia and macrophages polarize toward distinct phenotypes, with proinflammatory phenotypes persisting in the ischemic core, while anti-inflammatory/reparative phenotype cells accumulate in the peri-infarct regions.[37] Concurrently, adaptive immune responses peak, characterized by the temporal dynamics of T-cell subsets.[38] While limited T-cell infiltration may occur within 24 hours post-ischemia, substantial accumulation begins at day 3, with $CD8^+$ cytotoxic T cells peaking early (3–7 days) and regulatory T cells (Tregs) infiltrating later.[39] Tissue injury by cytotoxic $CD8^+$ T cells is mediated by the release of granzymes and perforin.[40] In contrast, Tregs' recruitment is delayed but ultimately cerebroprotective because Tregs produce

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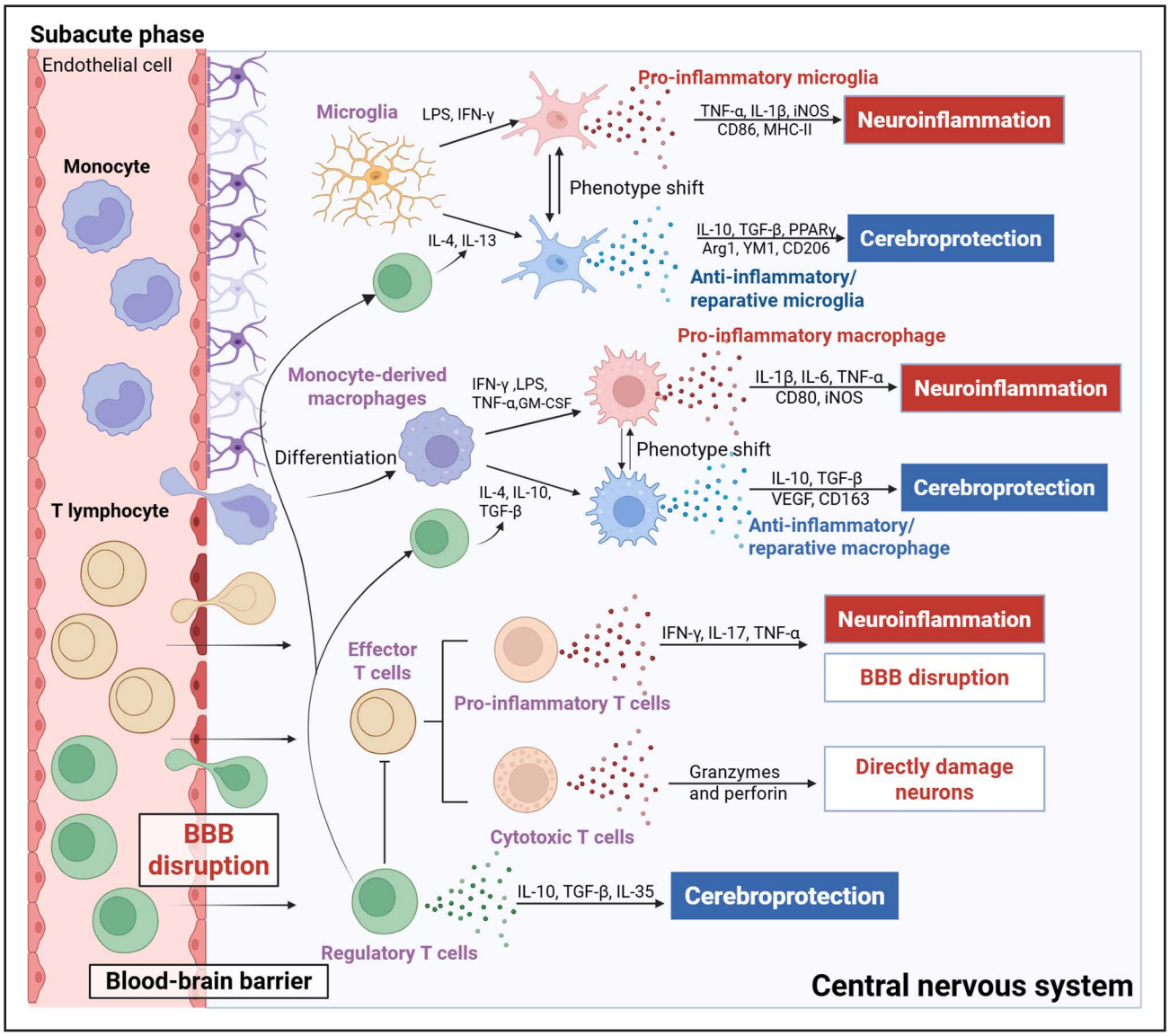

Figure 2. Immunologic dynamics in the subacute phase of ischemic stroke.
During the subacute phase of ischemic stroke, microglia and macrophages undergo distinct phenotypic polarization. Proinflammatory subsets perpetuate neuroinflammation and secondary injury, whereas anti-inflammatory/reparative phenotypes contribute to cerebroprotection. Concurrently, early infiltrating effector T cells (including proinflammatory and cytotoxic $CD8^+$ T cells) amplify neural damage through cytotoxic mechanisms. In contrast, regulatory T cells (Tregs) infiltrate at later stages and confer cerebroprotection by secreting anti-inflammatory cytokines such as IL (interleukin)-10 and TGF-β (transforming growth factor beta), as well as facilitating the shift of microglia/macrophages toward a reparative phenotype. Arg1 indicates arginase 1; GM-CSF, granulocyte-macrophage colony stimulating factor; IFN-γ, interferon-γ; iNOS, inducible nitric oxide synthase; LPS, lipopolysaccharide; MHC-II, major histocompatibility complex II; PPARγ, peroxisome proliferator-activated receptor γ; TNF-α, tumor necrosis factor alpha; VEGF, vascular endothelial growth factor; and YM1, active recombinant mouse YM1/chitinase 3-like3 protein. Created in BioRender. Yan, L. (2026) https://BioRender.com/z6is5bq.

anti-inflammatory cytokines such as IL-10 and TGF-β (transforming growth factor beta), which can promote the transition toward anti-inflammatory/reparative macrophage polarization.[41]

Opposing T-cell effects represent potentially actionable immunomodulatory targets during early stroke recovery,[39] which can impact long-term outcome. For example, chronic cognitive decline and network reorganization, which develops weeks to months after stroke,[42] might be mitigated by acute-phase interventions and postoperative immune modulation targeting cytotoxic immune mechanisms.[4]

The subacute phase presents a relatively long therapeutic window for inducing potentially beneficial actions in patients with EVT. An exciting therapeutic target would be the development of immune therapies that promote a shift in microglia/macrophage polarization toward anti-inflammatory and reparative phenotypes, which might improve post-EVT recovery and might favorably rebalance cytotoxic T-cell effects with Treg actions.

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Impact of EVT on the Peripheral Immune System in Patients With Ischemic Stroke

Mechanical Vascular Injury and Immune Disturbance During EVT Procedures

Stent retrieval technology has improved substantially, resulting in improved recanalization and fewer complications. However, in addition to reperfusion injury,[43] EVT mechanically stresses vascular walls and generates shear forces that damage the endothelial glycocalyx, exposing P-selectin and other adhesion molecules that promote neutrophil binding through PSGL-1 (P-selectin glycoprotein ligand-1) interactions.[44] Neutrophil binding, in turn, activates platelets via αIIbβ3 (integrin alpha-IIb beta-3) integrin signaling pathways, leading to the formation of proinflammatory microthrombi that contribute to distal microvascular obstruction.[45,46] Thus, the incidence of vascular injury following EVT increases from 37% to 61% with one catheter pass to 52% to 70% with a second pass.[47] Flow cytometric analysis of $CD31^+$ cells in intracranial blood samples distal to the stent retriever in 16 patients with ischemic stroke identified true endothelial cells in one patient,[48] who subsequently developed malignant hemorrhagic transformation. However, further studies are warranted to confirm this mechanism and generate clinically applicable evidence.

Innate Immunologic Consequences of Reperfusion Injury Following EVT

EVT can trigger postischemic reperfusion injury (Figure 3).[49] Beyond the classical oxygen-calcium overload and oxidative stress caused by ischemic reperfusion, abrupt reperfusion triggers fundamental immunologic changes. A key example is neutrophil extracellular trap formation. This process releases chromatin-derived circulating free DNA, which activates the AIM2 (absent in melanoma 2) inflammasome and exacerbates neuroinflammation.[50] Elevated circulating free DNA concentrations in plasma after thrombectomy strongly predict poor 90-day functional outcomes, suggesting that circulating free DNA quantification may be a prognostic biomarker for long-term stroke recovery.[51] Concurrently, ischemic injury and peripheral inflammation can increase MIF (macrophage migration inhibitory factor) levels, subsequently mediating endothelial cell death after reperfusion via the activation of RIPK1 (receptor-interacting protein kinase 1). The consequence is compromised BBB integrity, which leads to vasogenic edema and hemorrhagic transformation.[52]

Immunovascular Crosstalk in Microcirculatory Disorders Following EVT

Lack of reflow after apparently successful EVT macrovascular recanalization reflects complex vascular-immune interactions.[13] Synergistic interactions among 3 interrelated pathological mechanisms culminate in a no-reflow situation: (1) microthrombosis from endothelial injury, which triggers platelet aggregation, leukocyte adhesion, and neutrophil capillary obstruction[46,53]; (2) microvascular compression from cytotoxic edema via ionic imbalances and AQP4 (aquaporin-4) upregulation[54]; and (3) microvascular constriction from reactive oxygen species (ROS)-mediated pericyte contraction.[55] In both the EXTEND-IA (Extending the Time for Thrombolysis in Emergency Neurological Deficits-Intra-Arterial) and EXTEND-IA TNK (Tenecteplase Versus Alteplase Before Endovascular Therapy for Ischemic Stroke) EVT trials, regions of persistent parenchymal hypoperfusion were identified in a quarter of patients with EVT (33/130). Unsurprisingly, patients with parenchymal hypoperfusion experienced poor outcomes, hemorrhagic transformation, and infarct growth at 90 days.[56]

Successful recanalization does not guarantee immediate tissue reperfusion, as postthrombectomy hemodynamic impairment may persist due to immune-mediated mechanisms, underscoring the need for deeper mechanistic inquiry and the potential value of standard imaging in risk stratification.

General Anesthesia or Conscious Sedation for Patients With EVT?

General anesthesia is typically recommended for patients with impaired consciousness, agitation, or nausea/vomiting[12] because it provides immobilization and a secure airway. However, general anesthesia also requires a time-consuming induction during a period when delays in recanalization can worsen brain injury.[57] The advantages and disadvantages of general anesthesia and conscious sedation for EVT are shown in Table 1.[58,59]

Early observational studies associated general anesthesia with worse functional outcomes and higher mortality compared with procedural sedation during EVT[60] although findings were probably confounded by differences in baseline severity and variable anesthesia protocols.[11] Subsequent randomized trials implementing standardized hemodynamic and ventilatory management reported comparable outcomes between approaches, with some evidence favoring general anesthesia in protocolized settings.[61] The general anesthesia versus procedural sedation debate continues as optimal general anesthesia outcomes depend critically on both rapid drug availability and anesthesiologic expertise. With either approach, maintaining hemodynamic stability and minimizing reperfusion delays are decisive.[11]

Compared with general anesthesia, procedural sedation provides superior hemodynamic stability, reduced reperfusion delays, and the advantage of preserved neurological monitoring that can be adjusted according to sedation depth.[57] However, procedural sedation may fail to control patient agitation or movement, which can delay canalization and increase the risk of complications including vascular perforation and aspiration pneumonia.[57] Notably, intraoperative conversion to general anesthesia occurs in ≈12% of cases and is associated

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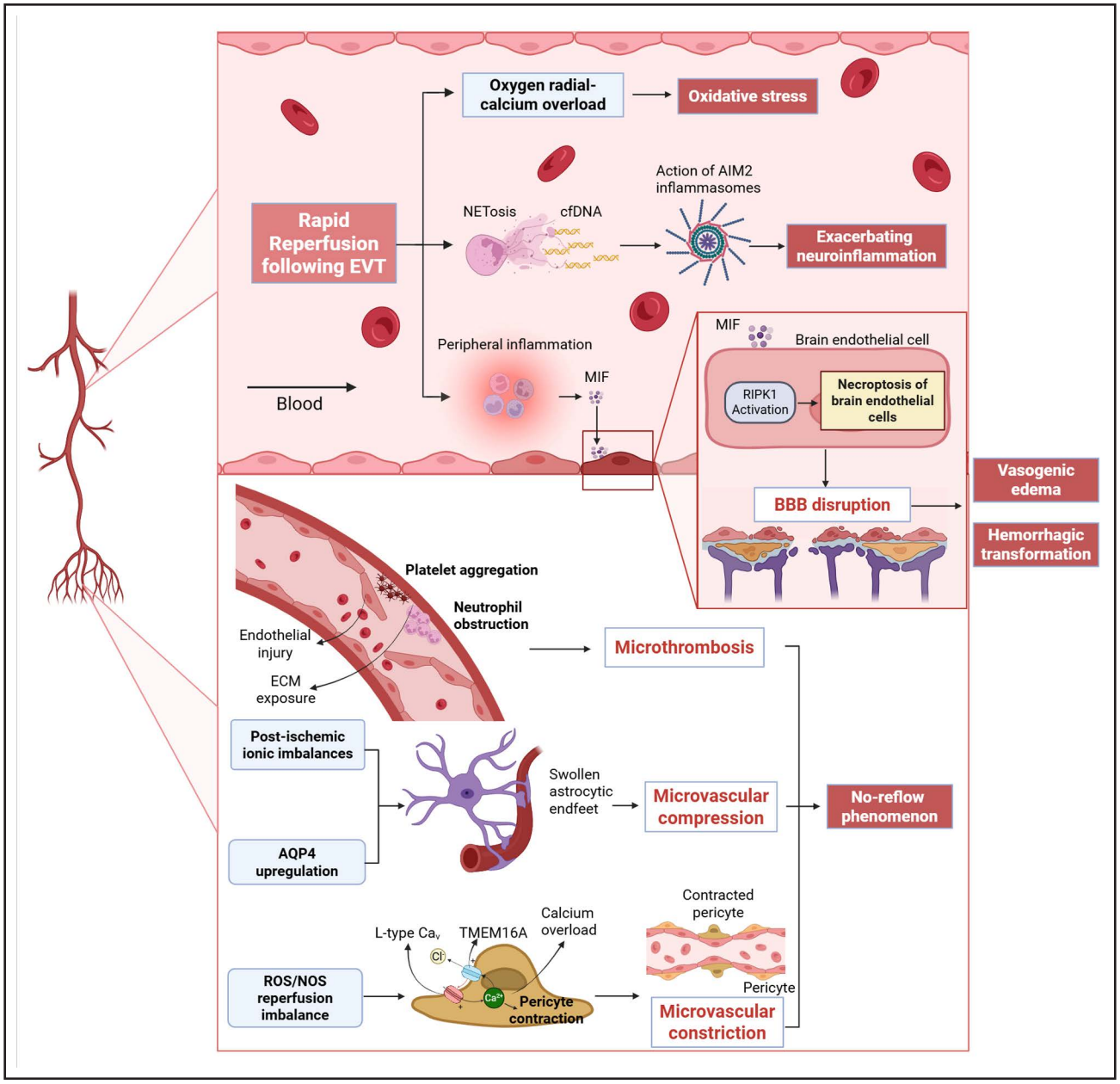

Figure 3. Integrated mechanisms of post-endovascular thrombectomy (EVT) reperfusion injury and immunovascular crosstalk. EVT triggers neutrophil extracellular trap formation (NETosis), releasing circulating free DNA (cfDNA) that activates the AIM2 (absent in melanoma 2) inflammasome and amplifies neuroinflammation. Concurrently, MIF (macrophage migration inhibitory factor) engages endothelial surface receptors, activating RIPK1 (receptor-interacting protein kinase 1)–mediated cell death, compromising blood-brain barrier (BBB) integrity, which, in turn, promotes vasogenic edema and hemorrhagic transformation. At the microcirculatory level, the no-reflow phenomenon arises from microthrombosis, compression by cytotoxic edema, and reactive oxygen species (ROS)-induced pericyte contraction. AQP4 indicates aquaporin-4; ECM, extracellular matrix; NOS, nitric oxide synthase; and TMEM16A, transmembrane protein16A. Created in BioRender. Yan, L. (2026) https://BioRender.com/49yq1fk.

with poorer prognosis.[62] Despite the persistent controversy, the latest American Heart Association guidelines endorse either anesthetic approach, provided that the decision is tailored to patient risk factors, procedural requirements, and clinical context (class II recommendation).[8] Based on available evidence, anesthetic selection should be individualized considering the patient condition and procedural requirements.

Preclinical Evidence for the Efficacy of Immunomodulation in Ischemic Stroke

Targeting Immune Signaling Pathways in Cerebral Ischemia

Regardless of the choice between general anesthesia and procedural sedation, current perioperative care prioritizes hemodynamic and respiratory stability. Animal

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Table 1. Advantages and Disadvantages of General Anesthesia Versus Procedural Sedation in EVT

	General anesthesia	Procedural sedation
Indications	Impaired consciousness	Conscious, cooperative patients
	Agitation	Supplemental sedation when local anesthesia is insufficient
	Nausea/vomiting	
Advantages	Complete immobilization	Better hemodynamic stability
	Secured airway management	Reduced reperfusion delays
	Pain-free procedure	Preserved neurological monitoring
	Potential higher reperfusion success with protocol-guided general anesthesia	Titratable sedation depth
Disadvantages	Time-consuming induction/intubation	Patient movement/agitation may prolong reperfusion
	Risk of hemodynamic instability	Complications (vascular perforation and aspiration pneumonia)
	Potential procedural delays	≈12% conversion to general anesthesia (associated with worse outcomes)
Key considerations	Optimal outcomes depend on rapid drug availability and an anesthesiologist's expertise	Requires careful patient selection to avoid intra-operative conversion to general anesthesia
	Hemodynamic stability and minimizing reperfusion delays are critical	

EVT indicates endovascular thrombectomy.

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studies, particularly those employing the transient middle cerebral artery occlusion model to mimic postthrombectomy reperfusion, suggest that targeted immunotherapy may improve outcomes of patients with stroke. However, clinical benefit remains entirely speculative.[63]

Immune signaling molecules, including cytokines, chemokines, and adhesion molecules, play pivotal roles in coordinating postischemic neuroinflammation.[63] Proinflammatory microglia-derived TNF-α induces endothelial necroptosis and BBB disruption during ischemia/reperfusion, with TNF-α inhibitors such as infliximab demonstrating cerebroprotective effects by reducing endothelial necroptosis and BBB disruption in rodent models.[64] The NLRP3 (NLR family pyrin domain-containing 3) inflammasome is activated by ischemic hypoxia, and the inhibition of its upstream activator purinergic receptor P2X7 (purinergic 2X7 receptor) is reported to reduce infarct volume and improve neurological outcomes.[65] RIPK1 also regulates endothelial necroptosis, and both its pharmacological inhibition (via necrostatin-1) and inhibition of its upstream regulatory factor MIF (via (S,R)-3-(4-hydroxyphenyl)-4,5-dihydro-5-isoxazole acetic acid methyl ester [ISO-1]) can attenuate ischemic brain damage.[52] IL-6 exhibits complex temporal dynamics, displaying both detrimental early effects and tendentially beneficial late-stage actions.[66] Other potential targets include selectins and integrins though clinical translation has proven challenging.[67]

Therapeutic Modulation of Microglia/Macrophage Phenotypes After Stroke

Microglia/macrophages can be polarized to proinflammatory phenotypes via classical activation pathways, which exacerbate brain injury or polarized to anti-inflammatory/reparative phenotypes through alternative activation pathways that might be cerebroprotective.[68] Anti-inflammatory microglia primarily secrete factors including IL-4, IL-10, and TGF-β, which suppress neuroinflammation. Multiple strategies targeting these factors have demonstrated efficacy in preclinical models.[69–71] Small molecule modulators such as PPARγ (peroxisome proliferator-activated receptor γ) agonists (rosiglitazone and pioglitazone)[72] and AMPK (AMP-activated protein kinase) activators (metformin) promote anti-inflammatory microglia polarization while suppressing proinflammatory responses.[73] Natural compounds and other pharmacological agents have shown complementary effects.[74] Whether any of these pathways and presumed mediators can be used clinically remains unknown.

Therapeutic Modulation of T-Cell Polarization Toward Anti-Inflammatory Th2 and Treg Phenotypes

Ischemic brain injury activates adaptive immune responses. For example, T helper cells (Th) differentiate into Th1, Th2, Th17, or Treg subsets depending on microenvironmental cues and interactions with innate immune cells. Given the anti-inflammatory effects of Th2 and Treg cells, polarization toward these 2 cell types holds therapeutic potential for poststroke immunomodulation.[75] Expansion of Tregs and Th2 polarization through IL-2/IL-2 antibody complexes or IL-33 administration shows promise in experimental stroke models.[41] Th2 polarization further promotes anti-inflammatory microenvironments through STAT6 (signal transducer and activator of transcription 6)-dependent pathways.[76] Treg/Th2-targeted immunomodulation is a promising strategy for stroke treatment, but Treg plasticity and optimal treatment timing require further investigation.

Other Promising Interventions That Exhibit Immune Modulatory Effect for Ischemic Stroke

Through paracrine signaling, stem cells attenuate neuroinflammation by suppressing proinflammatory cytokines and enhancing anti-inflammatory factors[77,78] while simultaneously promoting phenotypic conversion of microglia/macrophages from proinflammatory to anti-inflammatory phenotypes.[79] Mesenchymal stem cells further regulate the immune landscape by suppressing NK cell activity and promoting Treg infiltration.[80] In addition, stem cell–derived exosomes mediate cerebroprotection through miRNA transfer and other paracrine mechanisms.[81]

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Remote ischemic preconditioning represents a promising immunomodulatory strategy although the approach has failed in nearly every other context. A preclinical study reported that remote ischemic preconditioning increases the entry of monocytes into the injured brain, while infiltrating monocytes expressed higher CD36 levels, associated with CD36-dependent efferocytosis. This may rescue delayed transneuronal degeneration in the postischemic brain and promote stroke recovery.[82] Whether the approach will prove clinically useful remains highly speculative.

Table 2. Potential Immunomodulatory Targets in Ischemic Stroke Based on Integrated Preclinical Evidence

Therapeutic approach	Key targets	Potential interventions	Mechanisms of action	Preclinical evidence	References
Targeting immune signaling pathways	TNF-α	sTNFR, infliximab	Inducing endothelial necroptosis and BBB disruption via TNFR1 signaling	Cerebroprotection in rodent models	64
	NLRP3 inflammasome	NLRP3 inhibitors (eg, MCC950)	Mediating postischemic neuroinflammation and neuronal apoptosis via caspase-1 activation	Reduced infarct volume and neurological deficits	65
	P2X7R	P2X7R inhibitors	Regulating endothelial necroptosis and BBB dysfunction through TNFR1 and TLR4 pathways		
	RIPK1	RIPK1 inhibitors (eg, Nec-1)	Regulating endothelial necroptosis and BBB dysfunction through TNFR1 and TLR4 pathways	Attenuated ischemic brain injury	52
	MIF	MIF inhibitors (eg, ISO-1)			
	IL-6	IL-6R antagonist (eg, tocilizumab)	Dual role: early proinflammatory (BBB disruption) vs late neurotrophic (angiogenesis)	Neuroprotective effects in experimental ischemia	66
	Integrins (LFA-1 and Mac-1)	Integrin blockade	Facilitating leukocyte adhesion and transmigration, exacerbating neuroinflammation	Reduced neutrophil infiltration and infarct size	67
Anti-inflammatory microglia/macrophage polarization	IL-4/IL-10/TGF-β	IL-4/IL-33 administration	Promoting anti-inflammatory phenotype polarization via STAT6/Smad3, suppressing proinflammatory cytokines	Enhanced neuroprotection and tissue repair	69–71
	PPARγ	Thiazolidinediones (rosiglitazone)	Driving anti-inflammatory phenotype polarization by inhibiting NF-κB and activating Nrf2/ARE pathways	Reduced infarct volume via anti-inflammatory effects	72
	AMPK	AMPK activator (eg, metformin)	Inducing an anti-inflammatory phenotype via metabolic reprogramming and suppression of proinflammatory mediators	Improved functional recovery	73
Th2/Treg immunomodulation	IL-33/STAT6	IL-33 therapy	Expanding Treg and Th2 populations, enhancing anti-inflammatory cytokine production	Mitigated neuronal damage and inflammation	76
	IL-2	IL-2/IL-2 antibody complexes	Promoting Treg expansion and suppressing Th1/Th17 responses	Improved stroke outcomes	41
Stem cell–based therapy	MSCs	Intravenous MSC infusion	Paracrining suppression of proinflammatory cytokines and promotion of anti-inflammatory microglia polarization	Reduced neuroinflammation and enhanced neurorepair	80
	Stem cell–derived exosomes	Exosome administration	Transferring miRNA to inhibit microglial activation and astrogliosis	Preserved BBB integrity and reduced infarct size	81
Other promising interventions	Monocyte phenotype CD36 efferocytosis	Remote ischemic conditioning	Promoting proinflammatory monocyte infiltration and CD36-mediated phagocytosis, reducing neurodegeneration	Enhanced recovery, reduced neurodegeneration, and increased efferocytosis	82
	Epigenetic regulation	Pharmacological modulators	Dynamically shaping immune responses by altering chromatin accessibility and gene expression	Reduced neuroinflammation and improved recovery in robust models	83–86

AMPK indicates adenosine 5'-monophosphate (AMP)-activated protein kinase; ARE, antioxidant response element; BBB, blood-brain barrier; IL, interleukin; ISO-1, (S,R)-3-(4-hydroxyphenyl)-4,5-dihydro-5-isoxazole acetic acid methyl ester; LFA-1, lymphocyte function-associated antigen 1; Mac-1, macrophage-1 antigen; MIF, macrophage migration inhibitory factor; MSC, mesenchymal stem cell; Nec-1, necrostatin-1; NF-κB, nuclear factor-κB; NLRP3, NLR family pyrin domain-containing 3; Nrf2, nuclear factor-erythroid 2 related factor 2; PPARγ, peroxisome proliferator-activated receptor γ; P2X7R, P2X purinoceptor 7; RIPK1, receptor-interacting protein kinase 1; Smad3, SMAD family member 3; STAT6, signal transducer and activator of transcription 6; sTNFR, soluble TNFR; TGF-β, transforming growth factor beta; Th, T helper cell; TLR4, toll-like receptor 4; TNFR, tumor necrosis factor receptor; TNFR1, tumor necrosis factor receptor 1; TNF-α, tumor necrosis factor alpha; and Treg, regulatory T cell.

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Emerging evidence underscores the pivotal role of epigenetic regulation in poststroke immune modulation. The interplay of DNA methylation, histone modifications, and noncoding RNAs gene expression regulation fine-tunes the transcriptional landscape of immune cells, dynamically shaping neuroinflammation. Consequently, epigenetic targets offer a novel therapeutic avenue though their clinical translation poses significant challenges[83–86] (Table 2).

THE NEW CONCEPT OF IMMUNE MODULATION IN PATIENTS WITH ISCHEMIC STROKE

Several promising immunomodulatory targets identified in preclinical stroke research have advanced to early clinical evaluation but have yet to be incorporated into standard perioperative management.[4]

Recent clinical studies have explored various immunomodulatory approaches to improve outcomes in ischemic stroke, particularly in patients undergoing EVT.[4] Strategies can be broadly categorized into cytokine-targeted therapies, immune cell migration inhibitors, anti-inflammatory agents, and cell-based therapies.

Promising Immune Modulatory Strategies for Perioperative Treatment of EVT

IL-1 and IL-6 are the most intriguing cytokines that can be targeted for immune modulation after stroke. The IL-1ra (IL-1 receptor antagonist), a member of the IL-1 cytokine family, has shown promise in a phase IIa study (n=34). Administration of IL-1ra was correlated with a reduction in plasma inflammatory markers and was accompanied by improved neurological recovery compared with the control group.[87] Tocilizumab, an IL-6 receptor blocker, is currently under investigation in patients with EVT, with preliminary data supporting its potential cerebroprotective role.[88] The ICAM-1 (intercellular adhesion molecule-1) antibodies (enlimomab; an immune cell migration inhibitor) were suggested to protect against stroke. However, enlimomab increased both infarct volume and mortality due to complement-mediated neutrophil activation in a phase III trial,[89] illustrating the risks of nonspecific immunomodulatory effects.[90]

In recent years, various anti-inflammatory agents have been tested preclinically and in patients with stroke. Fingolimod, a sphingosine-1-phosphate receptor modulator that induces reversible lymphopenia by preventing lymphocyte egress from lymphoid organs, has clinically demonstrated benefits in reducing infarct growth and hemorrhagic transformation when combined with thrombolysis.[91] In contrast, natalizumab, an α4 integrin inhibitor,[39] yielded inconsistent results across trials. A phase II clinical investigation in 161 patients demonstrated significant enhancements in several predefined secondary and tertiary functional end points following,[92] but a subsequent phase IIb trial (n=277) failed to confirm benefit.[93] This inconsistency suggests that broadly targeting leukocyte migration may be insufficient to achieve clinical improvement because it can simultaneously suppress beneficial repair mechanisms.[93]

Minocycline, a tetracycline derivative with anti-inflammatory and cerebroprotective properties, has been associated with improved functional recovery in patients with stroke in 3 open-label trials,[94–96] leading to ongoing trials evaluating its adjunctive use in EVT.[97] Methylprednisolone, a synthetic glucocorticoid exerting broad-spectrum immunomodulatory effects through glucocorticoid receptor-mediated suppression of both innate and adaptive immune responses, was evaluated as an adjunctive therapy for patients with stroke undergoing EVT in the recently completed MARVEL trial (Methylprednisolone as Adjunctive to Endovascular Treatment for Acute Large Vessel Occlusion; n=1680).[98] While missing its primary end point, post hoc analyses showed a 3% reduction in hemorrhagic transformation risk and potential survival benefits.[98] Ongoing trials, MIRACLE (Methylprednisolone Adjunctive to Endovascular Thrombectomy for Stroke) and MATCH (Methylprednisolone With Endovascular Thrombectomy for Large Ischemic Stroke), are investigating its use in specific high-risk populations.[97]

Cell-based therapy represents a promising immunomodulatory approach for stroke treatment with the potential to replace damaged neural circuits while secreting growth and trophic factors and stimulating endogenous trophic factors to enhance intrinsic repair.[99] Cell-based approaches include bone marrow mononuclear cells,[100,101] multipotent adult progenitor cells, and MultiStem.[102,103] Each has demonstrated safety in clinical trials, but efficacy for improving stroke outcomes remains unclear. Challenges remain in optimizing delivery timing and patient selection to maximize therapeutic potential.

Diverse immunomodulatory strategies hold therapeutic potential (Table 3). Their progress, while facing translational challenges, paves the way for optimizing future approaches, particularly in target specificity, treatment timing, and personalization.

Practical Considerations to Develop Perioperative Immunomodulation Treatments for Patients With EVT

Practical implementation of immunomodulatory therapies in the context of EVT requires careful consideration of patient selection, timing, administration routes, and safety profiles.[4] From a theoretical perspective, patients with larger infarct volumes or anterior circulation occlusions may benefit most because of their intense inflammatory responses.[91]

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Table 3. Overview of Current Evidence and Applications of Stroke Immunology in Clinical Practice

Drug/intervention	Target/mechanisms of action	Trial	Outcomes	References
IL-1 receptor antagonist	IL-1 receptor antagonism	Phase II (ISRCTN74236229)	Decreased plasma markers of inflammation	87
Tocilizumab	IL-6 receptor blockade	IRIS trial	Under investigation	88
Fingolimod	S1P receptor modulation	Open-label trial	Reduced infarct volume, attenuated microvascular permeability, and decreased HT risk with tPA coadministration	91
		NCT04675762	Recruiting	97
Natalizumab	α4 integrin blockade	Phase II (NCT01955707)	Improved secondary and tertiary end points	92
		Phase IIb (NCT02730455)	Safe but no benefit	93
Enlimomab	Anti-ICAM-1	Phase III	Failure: increased mortality	89,90
		NCT05228080	Under investigation	97
Minocycline	MMP-9/PARP inhibition	Open-label clinical trials	Better neurological outcomes on day 90	94–96
		NCT05367362, NCT05487417, and NCT05487417	Under investigation	97
Methylprednisolone	Glucocorticoid receptor agonism	MARVEL trial (ChiCTR2100051729)	Secondary benefits only	98
		NCT06360458 and NCT06870448	Under investigation	97
BM-MNCs	Paracrine immunomodulation	Phase I/II (NCT00761982)	Safe and secondary benefits	100
		Phase II (NCT02178657)	Safe but no efficacy	101
Pluripotent adult progenitor cells	Multipotent stromal cell immunomodulatory effects	Phase II (NCT01436487)	Safe but no benefit	102
MultiStem	Multipotent cell effects	Phase II/III (NCT02961504)	Safe but no benefit	103

BM-MNC indicates bone marrow mononuclear cell; HT, hemorrhagic transformation; ICAM-1, intercellular adhesion molecule-1; IL, interleukin; IRIS, interleukin-6 receptor inhibition undergoing endovascular treatment in acute anterior circulation ischemic stroke; MARVEL, Methylprednisolone as Adjunctive to Endovascular Treatment for Acute Large Vessel Occlusion; MMP, matrix metalloproteinase; PARP, poly-ADP ribose polymerase; S1P, sphingosine-1-phosphate; and tPA, tissue-type plasminogen activator.

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The optimal therapeutic window for immunomodulatory interventions critically depends on the dynamic progression of poststroke immune responses, with current clinical protocols typically targeting the 6- to 72-hour timeframe to address early neuroinflammatory responses. However, the consistent failure of multiple trials to achieve primary end points suggests a significant disparity between theoretical potential and clinical efficacy, potentially attributable to suboptimal timing of single-dose administrations.[101] This therapeutic gap underscores the need for rigorous evaluation of time-dependent treatment effects across distinct poststroke phases. Furthermore, given the inherent complexity of stroke-induced immune dysregulation, monotherapeutic approaches may prove inherently limited, necessitating future exploration of combinatorial strategies to achieve meaningful clinical benefits.[93]

Safety remains paramount when repurposing existing immunotherapies or introducing novel agents, as evidenced by reports of progressive multifocal leukoencephalopathy and herpes encephalitis associated with multiple sclerosis drugs such as natalizumab and fingolimod.[104] These rare but fatal complications occur more frequently in patients with prior immunotherapy exposure, underscoring how treatment-induced impairment of immune surveillance may precipitate serious problems.[104]

PERSPECTIVES AND CONCLUSIONS

Immunomodulation may have the potential to improve outcomes in patients with stroke receiving EVT. However, clinical translation is hampered by the temporal dissociation between acute-phase intervention and the assessment of long-term functional recovery (eg, 90-day modified Rankin Scale), compounded by a lack

of biomarkers bridging this gap. This misalignment may obscure transient therapeutic benefits. Other critical challenges include identifying optimal therapeutic windows, potentially synergistic treatment combinations, and individualized therapeutic approaches. Whether general anesthesia or procedural sedation is preferable remains unclear although available evidence suggests that both are suitable in most patients with respect to procedural and neurological outcome. Preclinical investigations are urgently needed to elucidate how these anesthetic modalities influence immunomodulation, as this could be a critical variable affecting therapy outcomes.

Future progress hinges on innovative trial designs and the application of advanced molecular diagnostics including imaging to dynamically monitor immune responses. Interdisciplinary collaboration will facilitate the successful translation of potential immune therapies for EVT into clinical care, possibly optimizing neurological recovery and reducing poststroke disability. This may help to establish the rationale and evidence for clinical routine, targeted immunomodulation in patients with EVT. Clinicians should select general anesthesia versus sedation based on patient characteristics and individual preferences. Management should focus on providing good operating conditions and maintaining hemodynamic stability.

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ARTICLE INFORMATION

Affiliations

Department of Anesthesiology, Renji Hospital and Key Laboratory of Anesthesiology, Shanghai Jiao Tong University, Ministry of Education (L.C., Y. Li, W.Y., Y.Z., P.L.), Clinical Research Center, Renji Hospital (C.W., Y.Z., P.L.), and Department of Neurosurgery, Cerebrovascular Disease Center, Renji Hospital (J.W.), Shanghai Jiao Tong University School of Medicine, China. Department of Neurology, University Medical Center Hamburg-Eppendorf, Germany (T.M.). Department of Neurology, Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China (C.Y.). Department of Neurology, School of Medicine, University of Pittsburgh, PA (X.H.). Department of Neurology, University of Kentucky, Lexington (A.M.S.). Neurovascular Research Laboratory, Vall d'Hebron Institut de Recerca, Universitat Autònoma de Barcelona, Spain (A.R.). Sperling Center for Hemorrhagic Stroke Recovery, Burke Medical Research Institute, White Plains, NY (S.C.). Feil Family Brain and Mind Research Institute, Weill Medical College of Cornell University, New York, NY (S.C.). Faculty of Biology, Medicine and Health, Division of Neuroscience, School of Biological Sciences (S.M.A.) and Geoffrey Jefferson Brain Research Centre, Manchester Academic Health Science Centre, Northern Care Alliance NHS Foundation Trust (S.M.A.), The University of Manchester, United Kingdom. Department of Anesthesiology, The First Medical Center of Chinese PLA General Hospital, Beijing, China (Y.M.). Department of Anesthesiology, Xuanwu Hospital, Capital Medical University, Beijing, China (Y. Liu, T.W.). Center for Outcomes Research and Department of Anesthesiology, UTHealth, Houston, TX (D.I.S.). School of Life Sciences, University of Warwick, Coventry, United Kingdom (J.B.).

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